

$$\text{or } F = BvL = (28 \times 10^{-3})(25 \times 10^{-3})(12.88) = 9.0 \times 10^{-5} \text{ N}$$

- 7(a) Stopping potential is the **minimum** potential applied to the collector to stop even the **most energetic** electrons from the emitter from reaching the collector.

- (b)(i)1. gradient = h/e

$$h = \text{grad} \times e$$

$$= \frac{0.72 - 0}{(6.00 - 4.25) \times 10^{14}} \times 1.60 \times 10^{-19}$$

$$= 6.58 \times 10^{-34} \text{ J s}$$

Field Code Changed

2. From graph, threshold frequency, $f_0 = 4.25 \times 10^{14} \text{ Hz}$

Hence work function $\phi = h f_0$

$$= 6.58 \times 10^{-34} \times 4.25 \times 10^{14}$$

$$= 2.80 \times 10^{-19} \text{ J}$$

(ii) $f = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{694 \times 10^{-9}} = 4.32 \times 10^{14} \text{ Hz}$

Field Code Changed

From graph, at $f = 4.32 \times 10^{14} \text{ Hz}$, the $V_s = 0.02 \text{ V}$

- (iii) Below the threshold frequency of $4.25 \times 10^{14} \text{ Hz}$ which is the x-intercept (frequency-axis), no electrons will be emitted.
Hence the graph does not extend below the frequency axis.